

PREDICTION: LEGITIMATE

RISK SCORE: 16 / 100 (LOW RISK)

Scam Probability: 18.10% | Model Confidence: 81.90%

Banking Action: APPROVE

Official Recommendation: Approve Transaction. Transaction verified within normal customer behavior bounds. Instant settlement allowed.

1. Transaction Details & Actual Parameter Values

Transaction ID:	TXN-VERIFY-LEGIT-002	Transaction Amount:	\$2,500.00
Customer ID:	C123456789	Sender Old Balance:	\$10,000.00
Beneficiary ID:	M987654321	Sender New Balance:	\$7,500.00
Payment Type:	TRANSFER	Receiver Old Balance:	\$1,000.00
Device Type:	Mobile App	Receiver New Balance:	\$3,500.00
Location Region:	Domestic Home	6h Transaction Velocity:	2 txns
New Beneficiary:	No (Known)	Prior Fraud History:	No

2. Composite Risk Score Breakdown & Latency Timing

Risk Dimension	Sub-Score	Status & Impact
Model Fraud Probability	18.10%	Weight: 70% (Score: 18.1)
Amount Risk	0.0 pts	Transaction size evaluation
Velocity Risk	0.0 pts	6-hour transaction frequency
Device Risk	0.0 pts	Device profile verification
Location Risk	0.0 pts	Geographic regional risk
Beneficiary Risk	0.0 pts	New beneficiary check
Historical Behaviour	0.0 pts	Prior fraud flag check
Balance Behaviour	0.0 pts	Account wipeout detection
FINAL COMPOSITE RISK SCORE	16 / 100	LOW RISK CATEGORY

Feature Eng: 25.65 ms	Prediction: 1.22 ms	SHAP Engine: 1.66 ms	LIME Engine: 44.62 ms	Rendering: 1077.91 ms	TOTAL: 1151.65 ms
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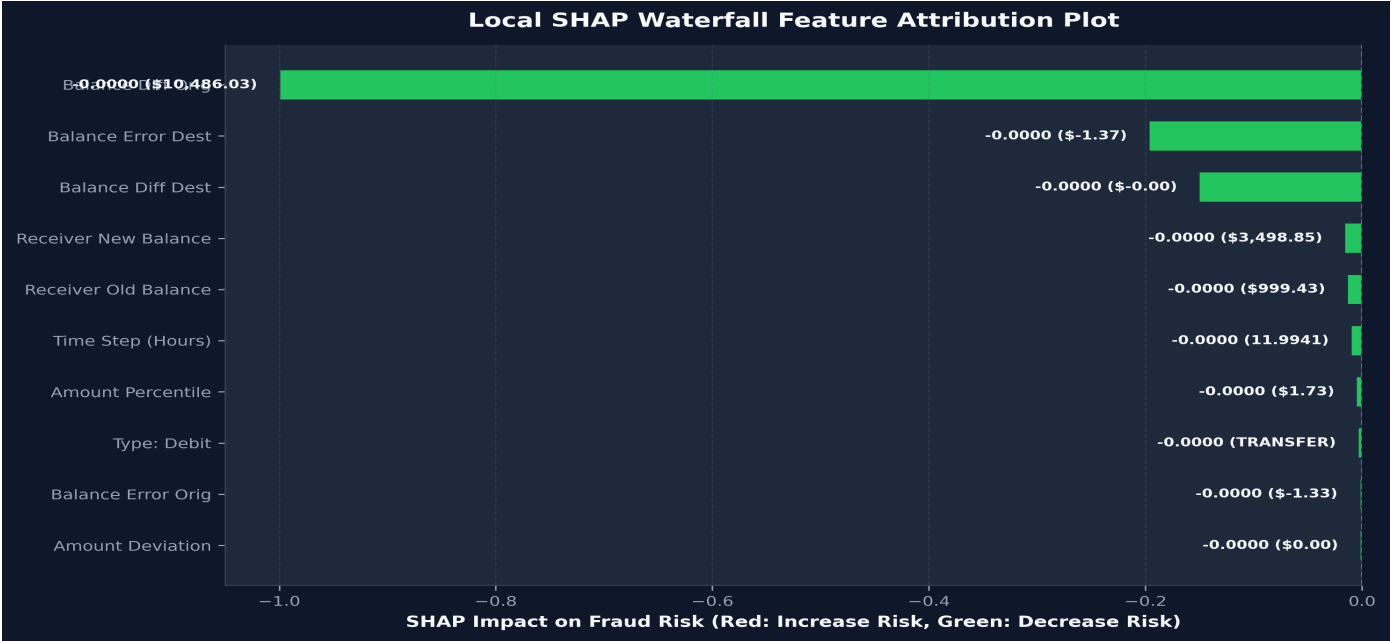
3. Banking-Grade Analyst Narrative Report

Transaction Summary:	Transaction TXN-VERIFY-LEGIT-002 involving Customer C123456789 transferring \$2,500.00 via TRANSFER to Beneficiary M987654321 on Mobile App from Domestic Home.
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Behaviour Analysis:	Sender account retained \$7,500.00 post-transaction. Normal transaction frequency: 2 recent transactions in 6 hours.
High Risk Indicators:	None identified.
Safe Indicators:	Standard Transfer Amount: \$2,500.00 is within routine spending range.; Account Balance Preserved: Remaining sender balance is \$7,500.00.; Normal Activity Velocity: Low frequency of 2 transactions in 6 hours.; Known Beneficiary: Beneficiary account has an established transfer history.; Recognized Customer Device: Access via registered customer mobile device.; Verified Domestic Location: Transaction initiated from primary home region.
Model Explanation:	GuidedGuard Gradient Boosting model evaluated this payment pattern with a scam probability of 18.10% and classified it as 'Legitimate'. The composite banking risk score is categorized under LOW RISK.
SHAP Summary:	Feature contributions align with baseline legitimate profile.
Final Recommendation:	Approve Transaction: Transaction verified within normal customer behavior bounds. Instant settlement allowed.

4. Top 10 SHAP Feature Contribution Attributions

Feature Name	Actual Value	SHAP Value	Contrib %	Direction	Impact Level
Balance Diff Orig	\$10,486.03	-0.0000	45.47%	Decreases Risk	High
Balance Error Dest	\$-1.37	-0.0000	8.95%	Decreases Risk	Medium
Balance Diff Dest	\$-0.00	-0.0000	6.86%	Decreases Risk	Medium
Receiver New Balance	\$3,498.85	-0.0000	0.75%	Decreases Risk	Low
Receiver Old Balance	\$999.43	-0.0000	0.62%	Decreases Risk	Low
Time Step (Hours)	11.9941	-0.0000	0.48%	Decreases Risk	Low
Amount Percentile	\$1.73	-0.0000	0.25%	Decreases Risk	Low
Type: Debit	TRANSFER	-0.0000	0.19%	Decreases Risk	Low
Balance Error Orig	\$-1.33	-0.0000	0.11%	Decreases Risk	Low
Amount Deviation	\$0.00	-0.0000	0.10%	Decreases Risk	Low



5. Local LIME Decision Rules & Explanations

Decision Rule Condition	Rule Weight	Effect Direction	Domain Explanation
Sender Old Balance = \$10,017.68	+1202.0613	Increases Risk	Feature condition 'Sender Old Balance = \$10,017.68' contributed positively to scam classification.
Sender New Balance = \$7,517.03	+901.9832	Increases Risk	Feature condition 'Sender New Balance = \$7,517.03' contributed positively to scam classification.
Receiver New Balance = \$3,498.85	+419.8026	Increases Risk	Feature condition 'Receiver New Balance = \$3,498.85' contributed positively to scam classification.
Transaction Amount = \$2,495.07	+299.3478	Increases Risk	Transaction amount is significantly elevated relative to normal threshold.
Receiver Old Balance = \$999.43	+119.8711	Increases Risk	Feature condition 'Receiver Old Balance = \$999.43' contributed positively to scam classification.
Time Step (Hours) = 11.9941	+1.3793	Increases Risk	Feature condition 'Time Step (Hours) = 11.9941' contributed positively to scam classification.
Type: Payment = TRANSFER	-0.0602	Decreases Risk	Payment type associated with routine account transfers.
Type: Cash Out = TRANSFER	-0.0597	Decreases Risk	Payment type associated with routine account transfers.
Type: Transfer = TRANSFER	+0.0597	Increases Risk	Transfer/Cash Out payment type has higher statistical fraud incidence.
Type: Debit = TRANSFER	-0.0595	Decreases Risk	Payment type associated with routine account transfers.

Local LIME Surrogate Decision Rules Explanation

